

Pre-training Buys Ordering, Not Calibration: An EveNet Data-Efficiency Test for κ_λ Density-Ratio Learning in $HH \rightarrow b\bar{b}\tau^+\tau^-$ (CMS Full Simulation)

Working note — $b\bar{b}\tau^+\tau^-$ κ_λ NSBI project

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Gates G0 / G0.5 of EVENET_FEASIBILITY_NOTE; 90-run sweep, branch `fm`

Abstract

We test whether the pre-trained event-level foundation model (FM) EveNet [1] improves the data efficiency of *density-ratio* learning for a Higgs self-coupling (κ_λ) measurement via neural simulation-based inference (NSBI), using CMS full-simulation $HH \rightarrow b\bar{b}\tau^+\tau^-$ events. Three arms — fine-tuned, frozen-backbone, and from-scratch — identical in data, architecture and head and differing *only* in initialisation, were trained at six training fractions $\times$ five seeds (90 runs) and evaluated on one fixed held-out split against two axes fixed before unblinding: discrimination (weighted area under the receiver-operating characteristic curve, AUC) and ratio calibration (integral and shape closure). The result is a clean dissociation. On discrimination, pre-training wins decisively and consistently: $+0.014$ – 0.017 AUC at every fraction $\geq 1\%$ (many standard deviations), equivalent to a ≈ 3 – $4.5\times$ data multiplier, with both pre-trained arms exceeding a feature-level gradient-boosted decision tree (BDT) ceiling. On calibration, pre-training buys *nothing*: no arm passes the closure gates at any fraction, the pre-registered adoption metric (closure left-shift) is undefined, and the arms trade places within seed errors. Seed-ensembling does not change this, because the closure offsets are same-sign systematic ($\mathbb{E}_{\text{ref}}[\hat{r}] - 1 \approx +0.13$ – 0.29) rather than zero-centred. The frozen arm matches fine-tuning everywhere, locating the value in the pre-trained *representation* rather than in backbone adaptation. We argue the calibration failure is a property of EveNet’s noise-conditioned classification objective rather than of the task, note that the feature BDT reaches $|\text{IC}| = 0.018$ where EveNet reaches 0.13 – 0.17 *despite worse AUC*, and propose a single decisive ablation.

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1 What was tested

Physics task. Learn per-event density ratios $r(x; \kappa_\lambda, \kappa'_\lambda)$ between κ_λ hypotheses for use in an unbinned NSBI likelihood. Discrimination between κ_λ points is driven by the m_{HH} lineshape and triangle–box interference, i.e. by a *continuous* deformation of one process’s spectrum rather than by separation of two distinct processes.

Data. CMS 2024 full-simulation $HH \rightarrow b\bar{b}\tau^+\tau^-$ Powheg samples, passed through the KIT CROWN analysis chain (published $\mu\tau_h/e\tau_h$ selection), converted to EveNet’s stock 7-feature token schema: four tokens per event (b_1, b_2, τ_h, ℓ), missing transverse energy in the global conditioning block, weights = `genWeight` $\times$ `puweight`. Requiring a valid $H \rightarrow b\bar{b}$ candidate drops 18.6–24.5% of events (κ_λ -dependent: softer $\kappa_\lambda = 5$ spectrum $\Rightarrow$ more failed second- b reconstruction). Class 0 = $\kappa_\lambda = 1$ (SM reference, 488k events), class 1 = $\kappa_\lambda = 5$ (339k). The merged, shuffled sample is split 80/10/10; **all** runs share one fixed 82.7k-event test split.

The contrast. Three arms differing only in initialisation:

finetune

20M-parameter pre-trained checkpoint, all weights trainable.

frozen

same checkpoint, backbone (GlobalEmbedding, PET, ObjectEncoder) frozen, heads trainable
— isolates the *representation* from optimiser effects.

scratch

random initialisation — the no-FM control.

Each at training fractions $\{0.003, 0.01, 0.02, 0.05, 0.1, 1.0\}$ ($\approx 2.5\text{k}$ to 830k events) $\times 5$ seeds = 90 runs. Options files are the authors’ own downstream recipes (fine-tune and from-scratch variants of the Exotic-Higgs study), vendored unchanged apart from the freeze flags.

External bar. A gradient-boosted decision tree on the 10 high-level features behind the published unbinned result, measured at full statistics on the same physics (Sec. 6).

2 Pre-registration record

The following were fixed *before* any sweep output existed, and are reproduced here unchanged:

- (P1) **Decision primacy (2026-07-12).** The adoption metric is the *closure left-shift*: the smallest training fraction at which an arm’s seed-mean passes both gates, compared between finetune and scratch; their ratio is the equivalent-data multiplier. A left-shift is an adoption-grade win

even at AUC parity; an AUC-only win with failing closure is *not* a win. AUC serves as a secondary no-regression guard.

- (P2) **Gates.** $|\text{IC}| \leq \max(1\%, 3 \times \text{stat})$ and stat-debiased shape $\text{RMS} \leq 0.05$ (= the analysis `CLOSURE_TOL`), with $\chi^2/\text{ndf} \approx 1$ indicating noise-only residuals.
- (P3) **Checkpoint selection (2026-07-14).** Every arm is evaluated at its best-validation-loss checkpoint (EveNet’s predictor otherwise loads the final, i.e. most overfit, epoch).
- (P4) **Fallback.** Should no arm pass the gates anywhere — leaving the left-shift undefined — the secondary reading is relative closure at matched fractions, reported as such.

The fallback (P4) is the branch that actually materialised.

3 Metrics

3.1 The closure identity

For scores p , labels y and weights w on the test split, the ratio estimate is $\hat{r} = \frac{p}{1-p} \cdot f_{\text{prior}}$, with f_{prior} the inverse of the class prior the training loss effectively used (measured, Sec. 8). Everything below rests on one identity. Average the *true* ratio over events drawn from the reference:

$$\mathbb{E}_{\text{ref}}[r] = \int p_{\text{ref}}(x) \frac{p_{\text{hyp}}(x)}{p_{\text{ref}}(x)} dx = \int p_{\text{hyp}}(x) dx = 1. \quad (1)$$

The reference density cancels against the denominator and what remains is a normalised probability density. So the true ratio, averaged over reference events, is *exactly* one — for any pair of hypotheses, with no free parameters. Intuitively, $r(x)$ says “this event is r times more likely under the hypothesis”; over the reference ensemble the regions where $r > 1$ and $r < 1$ must balance precisely. On a weighted Monte Carlo sample the expectation becomes the self-normalised weighted average $\sum_{i \in \text{ref}} w_i \hat{r}_i / \sum_{i \in \text{ref}} w_i$, which converges to Eq. (1) at rate $1/\sqrt{n_{\text{eff}}}$ — hence every closure number carries a statistical error and is “consistent with 1”, never “equal to 1”.

3.2 The two metrics

Integral closure, $\text{IC} = \mathbb{E}_{\text{ref}}[\hat{r}] - 1$

Definition: the weighted mean of $\hat{r}$ over held-out reference events, minus one; quoted with its weighted standard error. *Intuition:* reweight the reference sample by $\hat{r}$ and ask whether you recover the *total* hypothesis yield. $\text{IC} = +0.20$ means the reweighted reference overproduces the hypothesis by 20%. It is the *normalisation* or “exchange rate” error of the ratio. Two-bin illustration: reference (0.5, 0.5), hypothesis (0.8, 0.2) gives true $r = (1.6, 0.4)$ and $\mathbb{E}_{\text{ref}}[r] = 0.5(1.6) + 0.5(0.4) = 1$ exactly; a miscalibrated $\hat{r} = (2.0, 0.4)$ gives 1.2, i.e. $\text{IC} = +20\%$.

Shape closure, SC_{rms} and SC_{norm}

Definition: quantile-bin the score; in each bin compare the $\hat{r}$ -reweighted reference mass A_b with the actual hypothesis mass B_b , forming $\text{rel}_b = A_b/B_b - 1$; report the RMS over bins with each bin’s statistical variance $(1/n_{\text{eff},0} + 1/n_{\text{eff},1})$ subtracted in quadrature, so finite-sample noise does not masquerade as bias. *Intuition:* IC can be satisfied by errors that cancel globally — too large here, too small there. The shape test therefore applies Eq. (1) *differentially*: after reweighting the reference by $\hat{r}$, it must become statistically indistinguishable from the hypothesis sample *bin by bin*, which is precisely what “being the density ratio” means. We bin

in the classifier score because the score is monotone in $\hat{r}$ itself, so any miscalibration is forced to show up there.

We quote SC_{norm} alongside SC_{rms} : the same quantity after rescaling $\hat{r}$ by $1/\mathbb{E}_{\text{ref}}[\hat{r}]$, i.e. the shape error with the global normalisation divided out. This matters because a constant scale error shifts *every* bin equally and therefore contaminates SC_{rms} with the IC failure — the two raw numbers are not independent.

The decomposition matters because the two failure modes have different consequences: a global scale error in $\hat{r}$ is degenerate with the rate term of an NSBI fit and removable by construction, whereas a shape error deforms the per-event likelihood and is not. SC_{norm} is a *post-hoc* diagnostic added during analysis and is reported alongside, never instead of, the pre-registered gates. Validation: on synthetic data a pure $\times 1.4$ scale error gives $SC_{\text{rms}} = 0.399$ but $SC_{\text{norm}} = 0.000$, while a genuine temperature distortion gives 0.694 and 0.503 respectively.

What closure cannot catch. Both metrics test *self-consistency*, not *sufficiency*. A classifier trained on lossy features $\phi(x)$ converges to the exact ratio of the *marginalised* densities $p_{\text{hyp}}(\phi)/p_{\text{ref}}(\phi)$, which satisfies Eq. (1) identically and passes any test binned in a function of ϕ — the degenerate estimator $\hat{r} \equiv 1$ passes both metrics perfectly while carrying no information at all. Information loss does not appear as bias; it appears as a *flatter* ratio, i.e. as lost discrimination. Closure therefore catches lies, not omissions: it earns $\hat{r}$ the right to enter the likelihood, while the AUC axis (Sec. 4) is what shows the ratio is worth entering. This is also why a score-binned closure test cannot substitute for the analysis-level closure in physical observables, where the model is asked to reproduce distributions it was not free to choose the coordinates of.

4 Result 1 — discrimination (gate G0)

fraction	single model (seed mean $\pm$ s.d., $n = 5$)			5-seed ensemble		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.7427(57)	0.7498(80)	0.7573(122)	0.7576	0.7699	0.7738
0.01	0.7768(108)	0.7793(76)	0.7612(300)	0.7945	0.7935	0.7822
0.02	0.7941(43)	0.7931(80)	0.7804(19)	0.8050	0.8056	0.7877
0.05	0.8056(54)	0.8099(22)	0.7949(8)	0.8150	0.8167	0.8000
0.1	0.8163(22)	0.8169(9)	0.8003(25)	0.8214	0.8217	0.8075
1.0	0.8313(2)	0.8313(5)	0.8250(4)	0.8339	0.8338	0.8292

Table 1: Weighted AUC. Parenthesised digits are the seed standard deviation in units of the last decimals. The pre-trained arms lead at every fraction ≥ 0.01 ; only at 0.003 does the ordering invert.

Three readings:

1. **Pre-training wins, decisively.** At fractions 0.01–1.0 both pre-trained arms exceed scratch by 0.014–0.017 in AUC against seed standard deviations of 0.001–0.008 — many standard deviations, and monotone across the grid.
2. **The gain is worth ≈ 3 – $4.5\times$ the data.** Interpolating scratch’s curve logarithmically in fraction, finetune’s ensemble AUC at 0.01, 0.02, 0.05 and 0.1 requires scratch fractions of 0.033, 0.079, 0.222 and 0.437 respectively, i.e. multipliers $3.3\times$, $4.0\times$, $4.4\times$, $4.4\times$; at full statistics scratch never reaches finetune’s value at all.

3. **Frozen $\approx$ finetune everywhere** (differences ≤ 0.004 , mostly within seed errors). The benefit lives in the pre-trained *representation*; adapting the backbone adds nothing measurable. This is the sharpest single answer of the trial to the “did the pretext retain κ_λ -relevant continuous information?” question: it did.

The 0.003 inversion. At the smallest fraction scratch leads (ensemble 0.7738 vs. 0.7576/0.7699). That tier is in a pathological optimisation regime rather than a representational one: with $\approx 2.5k$ training events and a global batch of $\approx 8k$ across 4 GPUs, one “epoch” is a single gradient step, and the selected checkpoints sit at epochs 4–21, i.e. the models are compared after $\mathcal{O}(10)$ updates under *different* learning-rate recipes (the fine-tune recipe uses an aggressive head learning rate over a small body one; the scratch recipe is uniform). We regard this tier as a recipe artefact pending a batch-size/learning-rate ablation, not as evidence about representations.

5 Result 2 — calibration (gate G0.5)

fraction	IC (seed mean; stat. error ≈ 0.01 – 0.02)			SC _{norm} (seed mean $\pm$ s.d.)		
	finetune	frozen	scratch	finetune	frozen	scratch
0.003	0.130	0.468	0.202	0.318(66)	0.470(79)	0.348(109)
0.01	0.205	0.361	0.237	0.343(123)	0.337(160)	0.207(172)
0.02	0.292	0.206	0.205	0.273(81)	0.235(77)	0.170(56)
0.05	0.188	0.215	0.124	0.165(41)	0.195(46)	0.158(58)
0.1	0.229	0.272	0.332	0.164(33)	0.154(64)	0.165(69)
1.0	0.133	0.171	0.156	0.074(56)	0.044(44)	0.084(48)

Table 2: Closure, single models. Gates: $|IC| \leq \max(0.01, 3\sigma_{\text{stat}})$ and shape ≤ 0.05 . **No arm passes at any fraction.**

Verdict under (P1)/(P2): a null. The first-passing fraction is **none** for all three arms, so the closure left-shift — the pre-registered adoption metric — is undefined. Under the (P4) fallback, the arms overlap within seed errors and *trade places*: scratch is nominally best at 0.01–0.05 (largest gap at 0.02, $\approx 2.3\sigma$), the pre-trained arms at 0.1 and 1.0. There is no consistent ordering and no left-shift.

The failure is statistics-limited, not fixed. SC_{norm} falls monotonically with training fraction (0.32 $\rightarrow$ 0.07 finetune, 0.47 $\rightarrow$ 0.04 frozen, 0.35 $\rightarrow$ 0.08 scratch) and χ^2/ndf drops from 60–190 to 6–15. The estimator does converge; it simply has not reached the 1%/5% gates by 830k events in any arm.

Ensembling does not rescue it. Because $\mathbb{E}_{\text{ref}}$ is linear in r , the ensemble ratio $\bar{r} = \frac{1}{K} \sum_k r_k$ satisfies the identity $IC_{\text{ens}} = \frac{1}{K} \sum_k IC_k$ exactly: the ensemble carries no information beyond the *signed* mean of the members’ offsets. The measurement is unambiguous — in 15 of 18 cells the per-seed mean $|IC|$ and the signed mean are numerically identical (e.g. finetune 0.02: 0.2917 vs. 0.2917; frozen 1.0: 0.1706 vs. 0.1706), i.e. **all five seeds miss in the same direction**. The offsets are a same-sign systematic ($\mathbb{E}_{\text{ref}}[\hat{r}] - 1 \approx +0.13$ to $+0.29$), not zero-centred model noise, so there is nothing for averaging to cancel. (Ensembling does help in the three mixed-sign cells: finetune 0.003, 0.130 $\rightarrow$ 0.038; scratch 0.01, 0.237 $\rightarrow$ 0.047; scratch 0.05, 0.124 $\rightarrow$ 0.043.) Ensemble AUC improves slightly for all arms and preserves the FM advantage (Table 1).

6 The ceiling comparison

The feature-level BDT was measured on the same physics at full statistics with class-normalised weights (so its prior factor is unity and its closure numbers are directly comparable):

fraction	AUC	IC	shape
0.01	0.7917	0.202	0.391
0.1	0.8130	0.011	0.039
1.0	0.8174	0.018	0.000

Table 3: Kinematic ceiling (10 published features, gradient-boosted trees).

Two comparisons matter, and they point in opposite directions:

- **Discrimination: EveNet wins.** At 0.1 and 1.0 both pre-trained arms exceed the ceiling (0.821/0.822 vs. 0.813; 0.834 vs. 0.817), i.e. the object-level model extracts κ_λ information the 10 aggregate features discard. *Caveat:* the ceiling was measured on the un-cut flat-feature population, whereas the sweep applies the valid- b -pair requirement, so the sweep population is slightly easier; the margin at 1.0 (+0.017) is larger than we would expect this effect to explain, but the comparison carries an asterisk that the arm-to-arm comparisons do not.
- **Calibration: the BDT wins by an order of magnitude.** At full statistics the ceiling reaches $|\text{IC}| = 0.018$ and shape 0.000; every EveNet arm sits at $|\text{IC}| = 0.13\text{--}0.17$. *A simple boosted-tree ensemble with worse AUC produces a far better-calibrated ratio.* This is the observation that shifts the interpretation from “this task is hard to calibrate” to “this training objective does not calibrate” (Sec. 7).

7 Interpretation: a dissociation, and a candidate mechanism

The headline. Pre-training improved *ordering* substantially and *calibration* not at all. These are separable properties: AUC is invariant under any monotone remap of the score, and an entire family of classifiers shares a given AUC while only one member is the density ratio. A method evaluated solely by significance-improvement characteristic (SIC) or background rejection — as in the EveNet paper [1] and the controlled objective study [2] respectively — would report an unambiguous success here and would be structurally unable to see the calibration null. For NSBI, where the per-event ratio enters the likelihood, the calibration axis is the binding one.

Candidate mechanism (testable). EveNet’s classification head is trained under diffusion-time conditioning: the loss is $\alpha^2(t) \text{CE}(y, \hat{y})$ with a noising probability that our configuration — following the authors’ own downstream recipe — sets to 1.0, while prediction is performed at the clean endpoint. A t -averaged cross-entropy is *not* a proper scoring rule for the $t = 0$ posterior: its minimiser preserves the ranking of events (hence the strong AUC) while the logit *scale* need not equal that of the clean-input posterior (hence the systematic $\mathbb{E}_{\text{ref}}[\hat{r}] > 1$). Three observations are consistent with this and inconsistent with a residual convention error: the offsets are same-sign across seeds and arms *including scratch*; they vary cell-to-cell rather than being a common constant (a wrong prior factor would give an identical IC everywhere); and they correlate cell-by-cell with SC_{norm} (frozen 0.003: 0.47/0.31; frozen 1.0: 0.17/0.00), as expected if both are driven by per-event log-ratio noise — for which $\mathbb{E}_{\text{ref}}[\hat{r}] \approx e^{\sigma^2/2} > 1$, with $\mathbb{E}_{\text{ref}}[\hat{r}] = 1.2$ corresponding to $\sigma \approx 0.6$.

The decisive ablation. Re-run one column of the grid (say fraction 1.0, all three arms, 5 seeds = 15 runs, hours) with the classification noising disabled (`noise_prob: [0.0, 0.0]`), everything else identical. If $|\text{IC}|$ collapses toward the ceiling’s 0.018, the calibration null is a fixable property of the training recipe and the FM programme continues on a corrected objective. If it does not, the null is a property of the pre-trained representation’s transfer to continuous ratio estimation, and the objective-mismatch verdict of the feasibility note stands.

8 Corrections applied during analysis

For reproducibility, three defects were found and fixed *after* predictions existed but *before* the numbers above were read. All are estimator/plumbing fixes, not changes to the decision rule.

1. **Prior convention (largest effect).** The evaluator initially applied $f_{\text{prior}} = W_0/W_1 = 0.556$ (raw-weight training). Scanning the three candidate conventions on all 15 full-statistics runs selected *class-balanced* ($f = 1$) unanimously; EveNet equalises classes internally. The spurious factor had produced $|\text{IC}| \approx 0.30\text{--}0.47$ *flat in N* and inflated shape closure by the same constant (a scale error shifts every bin equally). This is why the corrected numbers, unlike the first pass, decrease with statistics.
2. **Weight treatment.** Closure now uses $|w|$: binary cross-entropy training cannot consume negative weights, so the reference density the model saw is the $|w|$ one (the reference sample carries 6.1% negative weights).
3. **Predict-path configuration.** Predict configs inherited a relative `options.yaml` path that resolved into the (empty) config farm, and lacked the head-inclusion block, so every prediction task died before loading a checkpoint. Both are now generated from the matching training config.

9 Caveats and scope

- **No sensitivity claim.** No σ_{κ_λ} or absolute-sensitivity statement is made or implied; the stock schema discards τ -identification and FastMTT $m_{\tau\tau}$ (equally in all arms), and the MC-statistics estimator question remains open upstream [3].
- **One channel, one hypothesis pair.** $\kappa_\lambda = 1$ vs. $\kappa_\lambda = 5$, $\mu\tau_h$ and $e\tau_h$ only. The $\kappa_\lambda = 0$ pair and the systematic-variation task (Money Plot 2, nominal vs. injected $\text{JES}(m_{HH})$) are untouched.
- **Population mismatch vs. the ceiling only** (Sec. 6); all arm-to-arm comparisons share identical events.
- **Five seeds** give seed errors of 0.03–0.17 on SC_{norm} , which is why the closure comparison is a null rather than a measured equality.
- **The 0.003 tier** is optimiser-transient-dominated (Sec. 4).
- One frozen cell (0.02, seed 3) and one scratch run (0.01, seed 2 — early-stopped at epoch 1) are degraded; both inflate their cells’ seed spread and neither changes a conclusion.

10 Implications and next steps

For the FM programme. G0 passes on discrimination with a 3–4.5 $\times$ data multiplier; G0.5 is a null. Under the pre-registered rule (P1) that combination is *not* adoption for NSBI — but it is also not the objective-mismatch verdict the feasibility note feared, because the representation demonstrably carries κ_λ -relevant information (frozen $\approx$ finetune, both above the ceiling on AUC). The programme’s decision therefore hinges on the ablation of Sec. 7, which is cheap.

Priorities.

- N1. [P0, S] Noising ablation at fraction 1.0 (15 runs) — decides whether the calibration null is recipe or representation.
- N2. [P1, S] G1 corner-encoding A/B (τ_h at the vacant flag corner): the NPZs already exist for both encodings.
- N3. [P1, M] Money Plot 2 (systematic-variation data efficiency): the FM’s low-statistics claim aimed at nuisance parameterisation, using the analytic injector where closure is exact by construction.
- N4. [P2, S] Low-fraction recipe ablation (batch size / learning rate at 0.003) to convert the inversion from artefact to measurement.
- N5. [P2, S] Re-measure the ceiling on the valid- b -pair population to remove the asterisk in Sec. 6.

For the EveNet authors (discussion, 16 July 2026). Three concrete items: (i) is a calibration-preserving classification mode available or planned — i.e. a $t = 0$ -only classification loss — given that the ranking transfers but the calibrated ratio does not? (ii) our G1 result will be the first data on reusing *vacant flag corners* for object types absent from pre-training (τ_h, γ); would they bless this as the zero-surgery interim, and (iii) would they consider retrofitting an explicit type-embedding table — initialised from the current flag patterns so it is function-preserving on day one — so that future object types become an append-a-row operation rather than schema surgery?

References

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